



IoT Project Proposal

Intelligent Standby Power Eliminator

CPE 445: Internet of Things

Submitted to: **Dr. Abbas Javed**

Submitted by:

Muhammad Faizan Shurjeel	FA22-BCE-012
Abdullah Laeeq	FA22-BCE-026

Contents

1	Overview	2
2	System Architecture	2
2.1	Cloud Layer	2
2.2	Edge Layer	2
3	Components	3
4	Features	3
5	Expected Outcomes	3
6	Future Enhancements	3

1 Overview

Many electronic devices consume "vampire power" when in standby mode, contributing to significant energy waste and higher electricity bills. Conventional smart plugs require manual intervention to cut power, which is often forgotten.

The proposed solution is a low-cost, IoT-based "Intelligent Standby Power Eliminator." This retrofit smart plug learns the power consumption patterns of an appliance. It automatically detects when a device has entered a low-power standby state for an extended period and completely cuts its power, eliminating energy waste without requiring user interaction.

2 System Architecture

The system uses a two-tier (Edge-Cloud) structure for real-time control and long-term analysis.

2.1 Cloud Layer

AWS IoT Core will be used on its free tier as the data ingestion and analytics backend. It will receive telemetry data from the edge device via MQTT and can be used to visualize long-term energy savings and consumption patterns.

2.2 Edge Layer

All real-time sensing and control logic resides at the edge for maximum responsiveness and offline functionality.

- **The ESP32 Smart Plug:** An ESP32 DevKit is integrated with an ACS712 non-invasive current sensor and a 5V relay.
 - The ACS712 continuously measures the real-time current drawn by the connected appliance.
 - The ESP32 runs an onboard algorithm that compares the measured power against user-defined thresholds for "Active" and "Standby" states.
 - If the device remains in a "Standby" state for a configurable duration (e.g., 10 minutes), the ESP32 triggers the relay to completely cut main power.
 - A simple web interface hosted on the ESP32 allows for initial configuration of the power thresholds.

Data Flow: The ESP32 at the edge continuously monitors power. After an automatic power-off event, or on a periodic basis, it sends a summary of power consumption (kWh) and energy saved to AWS IoT Core via MQTT. User commands are not needed for the core functionality.

3 Components

Main hardware components include an ESP32 DevKit V4, an ACS712 Hall Effect Current Sensor module, and a 5V Relay module. The estimated cost is under 2500 PKR.

Cloud components include AWS IoT Core on a free tier.

Software components include the Arduino IDE with the ESP32 core, FreeRTOS for task management, and the AWS IoT SDK for Embedded C.

4 Features

- Automatic, intelligent elimination of vampire power drain.
- Real-time current monitoring.
- User-configurable power thresholds via a local web interface.
- Cloud-based logging for long-term energy consumption and savings analysis.
- Low operational cost and inherently power-efficient design.

The edge-based control loop operates with millisecond latency. The cloud communication is asynchronous and does not affect the core functionality.

5 Expected Outcomes

A fully functional prototype of a smart plug capable of autonomously learning and acting on device power states to conserve energy. The project will demonstrate practical IoT integration, real-world sensor data processing, embedded decision-making at the edge, and cloud-based data analytics for energy management.

6 Future Enhancements

Possible additions include:

- **Machine Learning at the Edge:** Implement a tinyML model (e.g., using TensorFlow Lite for Microcontrollers) on the ESP32 to automatically learn and classify the Active/Standby states without requiring manual user configuration.
- **Scheduling:** Add time-based scheduling features.
- **Multi-Device Dashboard:** A central cloud dashboard to monitor and manage multiple smart plugs throughout a home.